Answer **one** question in this Section in the spaces provided.

7	(a)	State *Newton's second law of motion*.	
		.....	
		.....	
		.....	[2]
	L1	**The rate of change of momentum of a body is (directly) proportional to the resultant force acting on the body**	**B1**
		and is in the same direction as the resultant force.	**B1**
	(b)	A rollerblader approaches and glides up a smooth ramp of length 2.0 m along AB, inclined at an angle of 30° to the horizontal. The side view of this ramp is shown in Figure 7.1.	

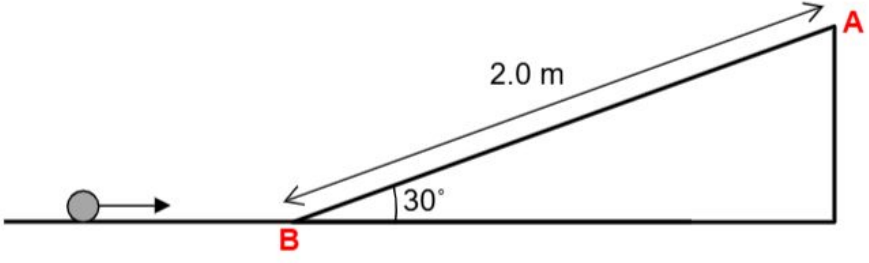


Figure 7.1

Figure 7.2 shows the ramp when viewed **perpendicular** to the ramp's inclined surface. The rollerblader glides up the ramp with initial speed u, at an angle of $\theta = 60°$ to the base of the ramp as shown in Figure 7.2. He stops exerting energy the moment he starts moving up the ramp.

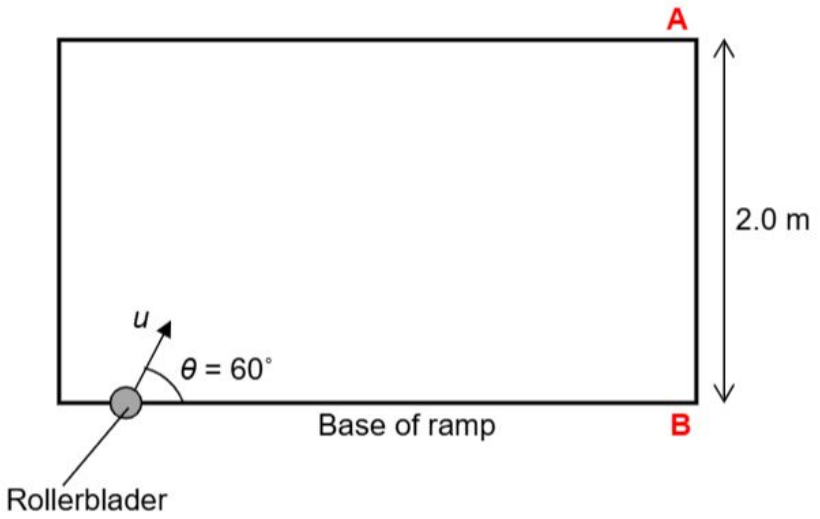
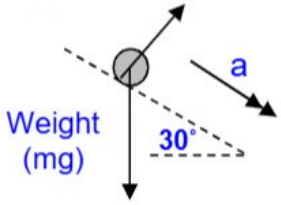


Figure 7.2

(i) Determine the magnitude of the acceleration of the rollerblader while moving up the ramp.

acceleration = ……m s^{-2} [2]

L1

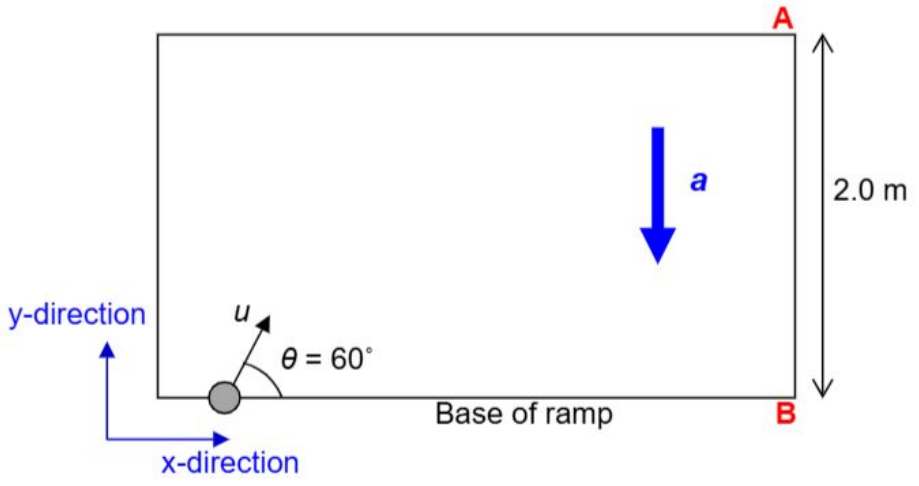


By Newton's 2nd law of motion,
mg sin30° = ma **M1**

a = g sin30° = (9.81) sin30°
= 4.905 = **4.9 m s^{-2} (or 4.91 m s^{-2})** **A1**

(ii) On Figure 7.2, draw an arrow to show the direction of the acceleration of the rollerblader. [1]

L1 Solution:



B1

(iii) Explain why the rollerblader does not move up the ramp along a straight path but along a parabolic path.

.....

.....

.....

.....

.....

..... [3]

L2

The acceleration is constant in magnitude and direction throughout the motion, and B1

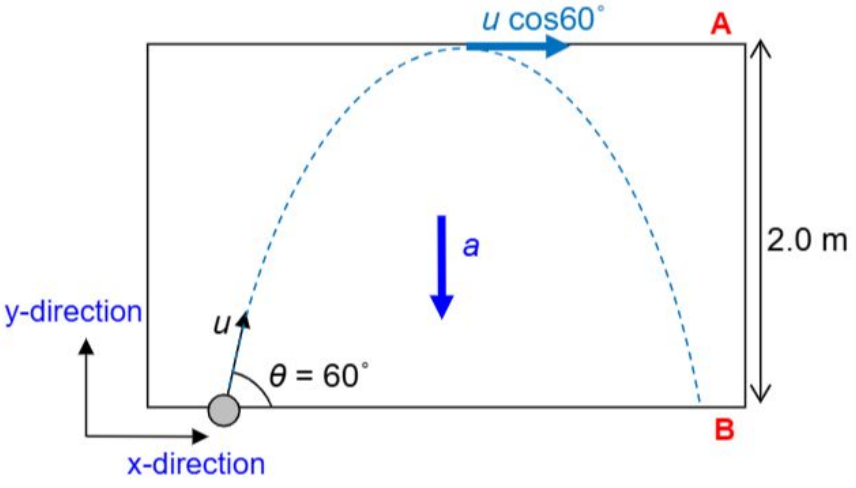
acts non-parallel to the velocity. B1

This causes the rollerblader to travel with uniform velocity in one direction (the x-direction) **and uniform acceleration in the perpendicular direction** (the y-direction)**.** B1

Hence the rollerblader traces a parabolic path.

(similar to projectile motion in a gravitational field when a particle is projected at an angle to the acceleration due to gravity)

(iv) By using the principle of conservation of energy, determine the minimum speed u for the rollerblader to just make it to the top of the ramp.

			minimum speed = ……..……m s^{-1}	[3]
		L3	Solution:  Loss in Kinetic Energy (KE) = Gain in Gravitational Potential Energy (GPE) $KE_i - KE_f = mgh$ **$\frac{1}{2} mu^2 - \frac{1}{2} m(u\cos60^\circ)^2 = mg(2.0 \sin30^\circ)$** $u^2 - (u\cos60^\circ)^2 = 2g(2.0 \sin30^\circ)$ $u = 5.1147 =$ **5.1 m s^{-1} (or 5.11 m s^{-1})** • ***M1 for correct expression for Change in KE_f.*** • ***M1 for correct expression for Change in GPE.*** • ***A1 for correct final answer.***	**M1** **M1** **A1**
		(v)	Using the speed in **(b)(iv)**, determine the time taken for the rollerblader to return to the base of the ramp.	

time taken = ...s [2]

L2 return to the base → $s_y = 0$

Take downslope as negative direction.

Use $s_y = u_y t + \frac{1}{2} a_y t^2$

$0 = (+5.1147\sin 60^\circ)\, t + \frac{1}{2} (-4.905)\, t^2$ **M1**

$t = 1.8061 =$ **1.8 s (or 1.81 s)** **A1**

(c) The ramp is extended to allow the rollerblader to move through a partial vertical circular motion with radius 3.0 m. Figure 7.3 shows the side view of the entire ramp. The rollerblader is upside-down at the top of the ramp.

The rollerblader now approaches the base of the ramp straight on perpendicularly.

You may regard the rollerblader as a point mass throughout this motion.

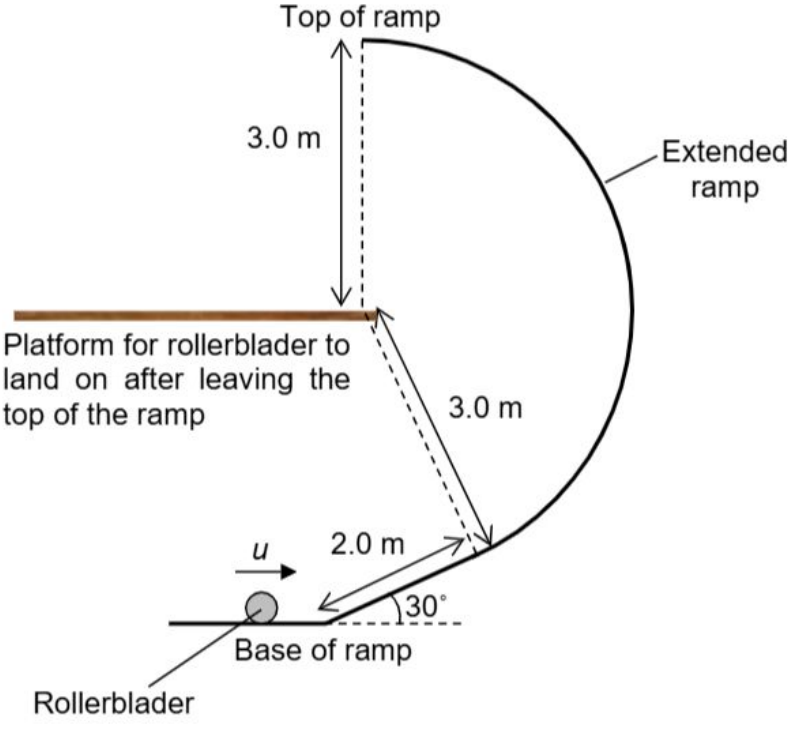


Figure 7.3 (not to scale)

(i) On Figure 7.4 below, sketch a graph to show how the magnitude of the centripetal acceleration a_c of the rollerblader varies with his linear speed v.

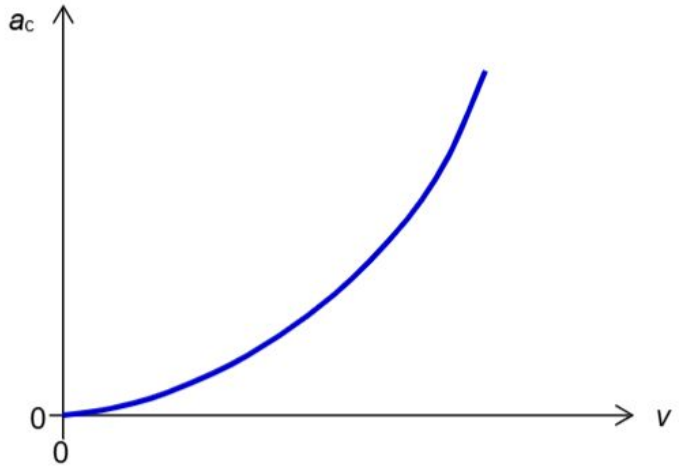


Figure 7.4 [1]

L2

$a_c = \frac{v^2}{r}$

Since r is constant, the graph is a **positive quadratic (parabolic shape).** **B1**

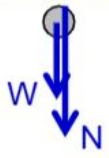


(ii) Determine the minimum speed *u* required for the rollerblader to reach the top of the circular ramp shown in Figure 7.3.

minimum speed = ……m s^{-1} [4]

L2 The minimum speed required occurs when the rollerblader *just* loses contact with the ramp at the top.

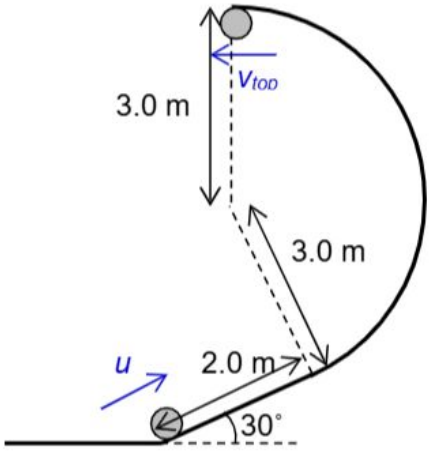
Free body diagram of the rollerblader at the top:



W: weight of rollerblader
N: normal contact force exerted by ramp on rollerblader

If he loses contact at this point, N = 0. **C1**

W = m v_{top}^2 / r
mg = m v_{top}^2 / r **M1**
v_{top} = √ (rg) = √ (3.0)(9.81)
= 5.4249 m s^{-1}



By conservation of energy,

$$\frac{1}{2}mu^2 = \frac{1}{2}mv_{top}^2 + mg[2.0sin30° + rcos30° + r]$$

$$\frac{1}{2}u^2 = \frac{1}{2}5.4249^2 + (9.81)[2.0sin30° + 3.0cos30° + 3.0]$$

M1

u = 12.605 = **13 m s^{-1} (or 12.6 m s^{-1})**

A1

(iii) On Figure 7.5 below, sketch a graph to show how the kinetic energy of the rollerblader E_K varies with his vertical displacement above the base of the ramp s_y when he moves up the ramp with this minimum speed.

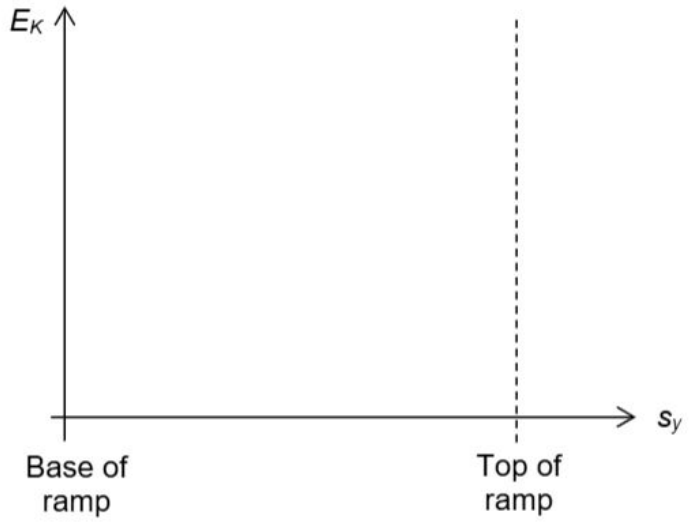
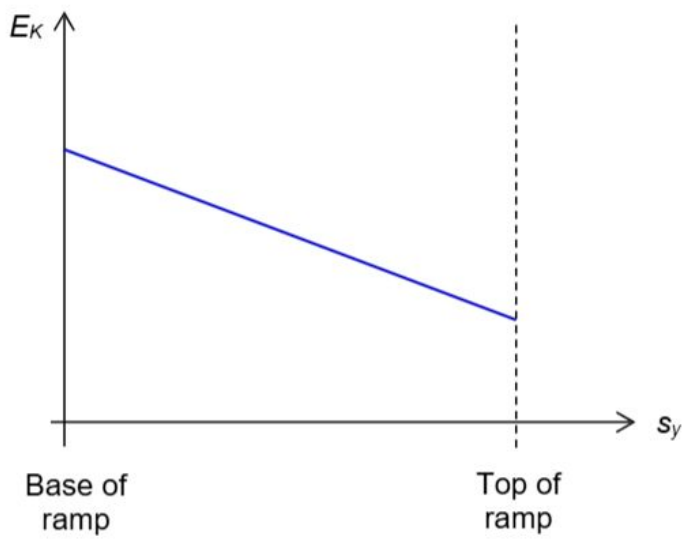


Figure 7.5 [2]

L3 GPE = mgs_y

$\Delta KE = -\Delta GPE$

Thus since ΔGPE varies linearly with s_y, ΔKE also varies linearly with s_y.



- **straight line, with negative gradient** B1
- **non-zero (since he moves up the ramp with the minimum speed, it means that his KE is always greater than zero)** B1

Straight line MUST be drawn using a ruler!

[Total: 20]